

1. Bibliometrics

As of April 1st 2025, the PI has published:

- 36 peer-reviewed original articles (8 peer-reviewed journal articles, 28 peer-reviewed conference articles), and 2 other articles.
- Citations: 555 (Google Scholar 2025-04-01)
- i10 index: 12 (Google Scholar 2025-04-01)

2. Selection of publications (reverse chronological order)

Dobslaw, Felix, and Feldt, Robert, and Yoo, Shin, and Yoon, Juyeon. "Challenges in Testing Large Language Model Based Software: A Faceted Taxonomy" *2025 Proceedings of Foundations of Software Engineering (FSE), 2030 Software Engineering - 2025*. ACM, 2025. (accepted for publication)

Contribution and Relevance: FD coordinated and led the collaboration. The DRT project is a natural continuation of this work, particularly regarding non-determinism in LLM testing in **A3**.

Jolak, Rodi, and Karlsson, Simon, and **Dobslaw, Felix**. "An empirical investigation of the impact of architectural smells on software maintainability." *Journal of Systems and Software* 225 (2025): 112382.

Contribution and Relevance: FD did supervision, co-writing and advisory as the most senior researcher. Highlights challenges of maintainability and particularly testability in often poorly structured projects, strengthening the need for DRT that focuses on isolatable carved-out units as dependency tests.

Dobslaw, Felix, and Feldt, Robert. "Automated Boundary Identification for Machine Learning Classifiers." *Proceedings of the 17th ACM/IEEE International Workshop on Search-Based and Fuzz Testing*. 2024.

Contribution and Relevance: FD led the work, and did most of the writing, and all of the experiments. Automatically identifying boundary behaviour for ML software includes correctness variance and confidence due to statistical non-determinism - a mandatory perspective for a generic formalization, as the non-determinism differs substantially from LLM produced.

Dobslaw, Felix, and Feldt, Robert, and Gomes de Oliveira Neto, Francisco. "Automated black-box boundary value detection." *PeerJ Computer Science* 9 (2023): e1625.

Contribution and Relevance: FD led the work, and did most of the writing and all of the experiments. Heuristic search to formalize/generalize and implement the first-ever process of automatically identifying boundaries for traditional software. Relevant as it required defining a complex formerly unmapped process, just like for DRT, with overlap in **A1**, **A2**, and **A4**.

Dobslaw, Felix, Ruiyuan Wan, and Hao, Yuechan. "Generic and industrial scale many-criteria regression test selection." *Journal of Systems and Software* 205 (2023): 111802.

Contribution and Relevance: FD did all the work - received help in terms of access to industrial data. Formalized the RT problem into a general framework using multi-objective search and

extraction of optimal bounds by integrating convex solving. Investigating it empirically with tailored search methods is relevant for the DRT project throughout all activities.

Holmberg, Mats, and **Dobslaw, Felix**. "An industrial case study on GUI testing with RPA." *2022 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*. IEEE, 2022.

Contribution and Relevance: FD supervised and guided student and co-authored. Relevant for the project as the industrial work with end-to-end tests revealed the many integration points that are implicitly tested in RT, but that are as of today not easily isolated when failures occur. DRT would lead to shorter fault-finding times in the industry.

Dobslaw, Felix, and Vallin, Morgan, and Sundström, Robin. "Free the Bugs: Disclosing Blocking Violations in Reactive Programming." *2020 IEEE 20th International Working Conference on Source Code Analysis and Manipulation (SCAM)*. IEEE, 2020.

Contribution and Relevance: FD supervised and guided student and co-authored. A deep understanding of state issues related to concurrent execution helped realise the value of DRT to assure that obtained failures are not due to dependency regressions, but due to within-project code that e.g. breaks a contract.

Dobslaw, Felix, and Feldt, Robert, and Michaëlsson, David, and Haar, Patrik, and Gomes de Oliveira Neto, Francisco, and Torkar, Richard. "Estimating return on investment for GUI test automation frameworks." *2019 IEEE 30th International Symposium on Software Reliability Engineering (ISSRE)*. IEEE, 2019.

Contribution and Relevance: FD led the work and coordinated the writing. The project gave a deep understanding of interfacing between industrial partners and new toolings and required not only technical sound implementations and evaluation, but included process thinking with the user in the loop.

Dobslaw, Felix, Tingting Zhang, and Mikael Gidlund. "QoS-aware cross-layer configuration for industrial wireless sensor networks." *IEEE Transactions on Industrial Informatics* 12.5 (2016): 1679-1691.

Contribution and Relevance: FD did all the work as a Phd student. Highly relevant to DRT, as this work expands SchedEx from [Dobslaw2014] by generalizing it to multiple layers in the ISO-OSI stack, i.e. routing and scheduling. The DRT project in **A3** has a similar challenge, to generalize the formalism from deterministic to non-deterministic.

Dobslaw, Felix, Tingting Zhang, and Mikael Gidlund. "End-to-end reliability-aware scheduling for wireless sensor networks." *IEEE Transactions on Industrial Informatics* 12.2 (2014): 758-767.

Contribution and Relevance: FD did all the work as a Phd student. The creation of SchedEx is highly relevant to DRT as a problem of arrival guarantees through scheduling in wireless sensor networks of varying topologies required to be formalized and solved for industrial use efficiently and effectively, and under the consideration of heavy non-determinism.

3. Publications 2017-2025

3.1 Original Articles

Jolak, Rodi, and Karlsson, Simon, and **Dobslaw, Felix**. "An empirical investigation of the impact of architectural smells on software maintainability." *Journal of Systems and Software* 225 (2025): 112382.

Thilderkvist, Eva, and **Dobslaw, Felix**. "On current limitations of online eye-tracking to study the visual processing of source code." *Information and Software Technology* 174 (2024): 107502.

Dobslaw, Felix, and Feldt, Robert, and Gomes de Oliveira Neto, Francisco. "Automated black-box boundary value detection." *PeerJ Computer Science* 9 (2023): e1625.

Dobslaw, Felix, Ruiyuan Wan, and Hao, Yuechan. "Generic and industrial scale many-criteria regression test selection." *Journal of Systems and Software* 205 (2023): 111802.

3.2 Conference Articles

Dobslaw, Felix, and Feldt, Robert, and Yoo, Shin, and Yoon, Juyeon. "Challenges in Testing Large Language Model Based Software: A Faceted Taxonomy" *2025 Proceedings of Foundations of Software Engineering (FSE), 2030 Software Engineering - 2025*. ACM, 2025. (accepted for publication)

Velander, Johanna, and Otero, Nuno, and **Dobslaw, Felix**, and Milrad, Marcelo. "Eliciting and Empowering Teachers' AI Literacy: The Devil is in the Detail." *International Conference in Methodologies and intelligent Systems for Technology Enhanced Learning* (pp. 138-152). Cham: Springer Nature Switzerland, 2024.

Gomes de Oliveira Neto, Francisco, and **Dobslaw, Felix**. "Building Collaborative Learning: Exploring Social Annotation in Introductory Programming." *Proceedings of the 46th International Conference on Software Engineering: Software Engineering Education and Training*. 2024.

Dobslaw, Felix, and Feldt, Robert. "Automated Boundary Identification for Machine Learning Classifiers." *Proceedings of the 17th ACM/IEEE International Workshop on Search-Based and Fuzz Testing*. 2024.

Rico, Sergio, and **Dobslaw, Felix**, and Öberg, Lena-Maria. "Starting collaborations between SMEs and researchers in software engineering." In *International Conference on Software Business* (pp. 222-230). Cham: Springer Nature Switzerland. 2024.

Haapasaari Lindgren, Marcus, Persson, Jon, Jolak, Rodi, and **Dobslaw, Felix**. "COMET: A ML-Based Tool for Evaluating the Effectiveness of Software Design Communication." *2023 ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)* (pp. 729-736). IEEE. 2023.

Dobslaw, Felix, and Peter Bergh. "Experiences with remote examination formats in light of GPT-4." *Proceedings of the 5th European Conference on Software Engineering Education*. 2023.

Dobslaw, Felix, and Angelin, Kristian, and Öberg, Lena-Maria, and Ahmad, Awais. "The gap between higher education and the software industry—A case study on technology differences." *Proceedings of the 5th European Conference on Software Engineering Education*, 2023.

Dobslaw, Felix, and Feldt, Robert. "Similarities of Testing Programmed and Learnt Software." *2023 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*. IEEE, 2023.

Cvitic, Petya Hristova, and **Dobslaw, Felix**, and Gomes de Oliveira Neto, Francisco. "Investigating software testing and maintenance of open-source distributed ledger." *2023 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*. IEEE, 2023.

Holmberg, Mats, and **Dobslaw, Felix**. "An industrial case-study on GUI testing with RPA." *2022 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*. IEEE, 2022.

Hokka, Heidi, and **Dobslaw, Felix**, and Jonathan Bengtsson. "Linking developer experience to coding style in open-source repositories." *2021 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*. IEEE, 2021.

Åhs, Fredrik, and Mozelius, Peter, and **Dobslaw, Felix**. "Artificial intelligence supported cognitive behavioral therapy for treatment of speech anxiety in virtual reality environments." *ECIAIR 2020*. Vol. 2. Academic Conferences and Publishing International Limited, 2020.

Dobslaw, Felix, and Gomes de Oliveira Neto, Francisco, and Feldt, Robert. "Boundary value exploration for software analysis." *2020 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*. IEEE, 2020.

Gomes de Oliveira Neto, Francisco, and **Dobslaw, Felix**, and Feldt, Robert. "Using mutation testing to measure behavioural test diversity." *2020 IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*. IEEE, 2020.

Dobslaw, Felix, and Vallin, Morgan, and Sundström, Robin. "Free the Bugs: Disclosing Blocking Violations in Reactive Programming." *2020 IEEE 20th International Working Conference on Source Code Analysis and Manipulation (SCAM)*. IEEE, 2020.

Dobslaw, Felix, and Feldt, Robert, and Michaëlsson, David, and Haar, Patrik, and Gomes de Oliveira Neto, Francisco, and Torkar, Richard. "Estimating return on investment for GUI test

automation frameworks." *2019 IEEE 30th International Symposium on Software Reliability Engineering (ISSRE)*. IEEE, 2019.

Feldt, Robert, and **Dobslaw, Felix**. "Towards automated boundary value testing with program derivatives and search." *Search-Based Software Engineering: 11th International Symposium, SSBSE 2019, Tallinn, Estonia, August 31–September 1, 2019, Proceedings 11*. Springer International Publishing, 2019.